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Metabolic status in mothers with gestational diabetes mellitus and the effects on fetal and neonatal outcomes – a cohort study

Helene Højbjerre[1], Sine Knorr[1,2], Lene Ring Madsen[1,3,4], Magnus Leth-Møller[1,2,8], Pernille Bundgaard Grinderslev[1], Amalie Nørkjær Svendsen[7], Laura Løftgaard Knudsen[1], Lene Sundahl Mortensen[5], Lars Peter Sørensen[6], Henrik Holm Thomsen[2,7], Per Glud Ovesen[1,2,8], Jens Fuglsang[1,2,8] and Ulla Kampmann[1,2*]

Abstract

Background Research on gestational diabetes mellitus has mainly focused on glucose metabolism, but improving glycemic control has not eliminated the complication rate in these pregnancies. The aim of this study was therefore to investigate how other metabolic and cardiovascular risk factors in women with gestational diabetes mellitus contribute to adverse pregnancy outcomes, thereby giving a more nuanced characterization of maternal metabolic status and the effects on fetal outcome.

Methods This cohort study included mother-infant pairs with maternal gestational diabetes mellitus ($n = 722$) in the Central Region of Denmark from 2019 to 2022. Anthropometric and metabolic data were collected during a pregnancy visit at week 34–38. Offspring data were collected from an audit of the participants' medical records. Large for gestational age (LGA) was defined as birth weight above the 90th centile for gestational age and sex.

Logistic and linear regression were performed to determine the association between maternal metabolic cardiovascular risk factors and adverse pregnancy outcomes. Analyses were adjusted for age, ethnicity, and parity. Exposures unrelated to weight were adjusted for pregestational body mass index. Furthermore, outcome rates were stratified by treatment (insulin vs. diet-only).

Results This study found increased odds for LGA with pre-pregnancy body mass index ≥ 30 kg/m^2 (adjusted odds ratio (aOR)) = 2.26 [1.37;3.73]), increasing diagnostic oral glucose tolerance test 2-h glucose value (aOR = 1.20 [1.05;1.38]), 3rd trimester total cholesterol (aOR = 1.25 [1.05;1.49]), 3rd trimester triglycerides (aOR = 1.28 [1.07;1.53]), 3rd trimester HbA1c (aOR = 1.13 [1.07;1.18]), high gestational weight gain (aOR = 1.76 [1.07;2.87]), and insulin-treatment compared to diet-only treatment (risk ratio = 2.51[1.89;3.33]).

Conclusion Strong associations between pre-pregnancy body mass index, glycemic measures, and lipid profiles with adverse neonatal outcomes, particularly large for gestational age, underscore the importance of a broader risk assessment in women with gestational diabetes mellitus, not only based on glycemic parameters but also including weight prior to and during pregnancy and lipid status.

*Correspondence:
Ulla Kampmann
ulla.kampmann@clin.au.dk

Full list of author information is available at the end of the article

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Keywords Gestational diabetes mellitus, Metabolism, Lipid status, Offspring, Gestational weight gain

Introduction

Gestational Diabetes Mellitus (GDM) is characterised by the development of hyperglycemia during pregnancy, usually followed by remission postpartum. Its global prevalence is increasing, posing a significant public health challenge [1–3]. GDM is associated with a range of adverse maternal and neonatal outcomes, including gestational hypertension/preeclampsia, preterm birth, shoulder dystocia, caesarean section (C-section), large-for-gestational-age (LGA, birthweight > 90th centile for gestational age and sex) infants, neonatal hypoglycemia, jaundice, respiratory distress syndrome (RDS), and neonatal intensive care unit (NICU) admission [1, 4]. Maternal glycemic status significantly affects GDM pregnancy outcomes [5–7], and enhanced maternal glycemic control notably reduces complication rates in GDM pregnancies; however, complication rates remain higher than those in normal pregnancies [4, 8]. The standard treatment for GDM in Denmark involves dietary and exercise interventions, with insulin treatment started if glycaemic control declines. Recent research has highlighted the heterogeneity in the physiological mechanisms driving hyperglycemia in women with GDM, indicating that the risk of adverse outcomes may vary according to the woman's metabolic profile [9–11]. Additionally, population demographics are shifting, with increasing maternal body mass index (BMI) and age contributing to a more complex clinical picture [12, 13]. Current studies have found a possible link between maternal lipid metabolism and fetal growth, though additional research is needed on maternal lipids and fetal outcomes [8, 14, 15].

The increasing GDM prevalence and the fact that complication rates in GDM pregnancies, including LGA babies, are still high despite more optimal glycemic control, indicate that the approach to GDM might have been too "glucocentric". This study aimed to investigate how metabolic and cardiovascular risk factors in a Danish cohort of mother-infant pairs with maternal GDM contribute to adverse obstetric and perinatal outcomes. By investigating different metabolic contributors and their associated effects on adverse pregnancy outcomes, an enhanced clinical approach in risk stratification for women with GDM will be obtainable.

Materials and methods

Study design and population

This prospective cohort study was conducted as part of a universal clinical follow-up program aiming to support the prevention of type 2 diabetes (T2DM) in women diagnosed with GDM in the Central Denmark Region from April 2019 to December 2022 [16]. This study enrolled only mothers with singleton pregnancies, and each participant was included only once. In Denmark, a risk-based screening program is used to detect cases of GDM. The diagnosis is made after 8 h of fasting using a 75 g oral glucose tolerance test (OGTT) in gestational week (GW) 24–28, and a 2-h value ≥ 9 mmol/l is diagnostic for GDM. Women diagnosed with GDM had standard clinical care visits at their local Department of Obstetrics and Gynaecology. Glucose monitoring included both pre- and postprandial measurements 6 times a day. According to Danish guidelines, treatment targets were 4–6 mmol/L preprandial and 4–8 mmol/L postprandial. Insulin treatment was initiated if two elevated plasma glucose levels, not explained by a single significant carbohydrate load, were measured over two weeks. Insulin treatment was therefore considered an indicator of more severe GDM. Insulin used in this cohort consisted of a combination of short-acting and intermediate-acting insulin (Novomix 30). During GW 34–38, blood and urine samples were collected, and blood pressure (BP), height, and weight were measured. Participants also completed a questionnaire providing background information.

Study population

All women diagnosed with GDM in the Central Denmark Region were invited to participate in the follow-up program.

Data collection

Data were extracted from medical records and collected via REDCap [17, 18]. Data were collected at five hospitals in the Central Denmark Region using standardised methods. Blood samples at GW 34–38 visits were non-fasting and analysed using routine hospital laboratory methods.

Characteristic

Maternal characteristics included age, parity, ethnicity, education, gestational hypertension and/or preeclampsia (PE), GDM treatment (insulin or diet-only), and the corresponding maximum dose of insulin (insulin max dose). Parity was categorised as nullipara (first pregnancy) or multipara (second or more pregnancies). Ethnic categories included Europe, Asia, the Middle East, and other categories. Educational level was grouped as follows: elementary school (10–11 years); short-term education (including upper secondary education, academy profession degree, and vocational education); intermediate higher education (professional bachelor's education, e.g., nurse or teacher); long higher education (e.g., master's or PhD); and other education (including foreign education). Gestational hypertension was defined in accordance with

The International Society for the Study of Hypertension in Pregnancy (ISSHP) and The International Association of the Diabetes and Pregnancy Study Groups (IADPSG) as a recurrent elevated BP, with an average of at least two measurements (≥ 140 mmHg systolic or/and ≥ 90 mmHg diastolic pressure) on at least two occasions, with onset GW ≥ 20. PE included proteinuria and/or organ failure alongside gestational hypertension [19, 20]. The insulin max dose was the highest insulin dose in International Units (IU) per kg pr day, based on the weight from GW 34–38 visit. Additional characteristics included gestational age (GA) at birth, and method of delivery, categorized as vaginal birth, vacuum-assisted delivery, and elective or emergency C-section. Neonatal characteristics included birthweight (BW), BW z-score, and sex.

Exposure variables

The pre-pregnancy BMI, diagnostic OGTT 2-h value, the metabolic profile at GW 34–38 (total cholesterol, triglyceride, low-density lipoprotein cholesterol (LDL-C), and haemoglobin A1c (HbA1c)), and gestational weight gain (GWG) were evaluated. Pre-pregnancy weight was self-reported and documented by the clinicians in the medical records and was classified as BMI < 25 kg/m^2, BMI ≥ 25- < 30 kg/m^2, and BMI ≥ 30 kg/m^2. GWG was calculated as the difference in kg between pre-pregnancy weight and weight measured at the GW 34–38 visit and classified per the Institute of Medicine (IOM) 2009 guidelines into recommended GWG, high GWG, and low GWG [21].

Outcome variables

Outcomes included preterm birth (before GW 37), shoulder dystocia, LGA, small-for-gestational-age (SGA), hypoglycemia, jaundice requiring treatment, RDS, and NICU admission lasting more than 24 h during the initial hospitalization after birth (NICU > 24 h) [20]. BW z-score was calculated using Scandinavian standards from Marsal et al., based on BW standardised to GA and sex [22]. LGA was defined as a BW z-score above 1.282, corresponding to the 90th percentile, while SGA was defined as a BW z-score below −1.282, corresponding to the 10th percentile of the standard population [20]. Neonatal hypoglycemia was defined as a blood glucose ≤ 2.5 mmol/L within 2 h of birth. In Denmark, the standard clinical practice involves early feeding and measuring blood glucose within 2 h in infants born to mothers with GDM, and a value ≤ 2.5 mmol/L defines hypoglycemia [23]. Definitions of shoulder dystocia, jaundice requiring treatment, and RDS were based on IADPSG outcome definitions [20].

Statistics

Maternal and neonatal characteristics are presented as mean and standard deviation (SD) for continuous variables with a normal distribution assessed by a quantile–quantile plot of residuals and as median and interquartile range (IQR) if a skewed distribution. Categorical variables are presented as counts n (%). No imputation was made for missing data; it was noted if missing data exceeded 10%.

The relationships between obstetric/perinatal outcomes and maternal metabolic variables were examined using logistic and linear regression, adjusted for maternal age, ethnicity, and parity. Regressions with diagnostic OGTT 2-h value, 3rd trimester total cholesterol, triglyceride, LDL-C, and HbA1c were further adjusted for pre-pregnancy BMI. Logistic regression results are reported as odds ratio (OR) with 95% confidence interval (95%CI), and it has been ensured that the data met the statistical assumptions. Consequently, the exposure variables pre-pregnancy BMI, GWG, and LDL-C were categorised as the assumption of a linear relationship with the model's log odds was not met. Furthermore, in the regression between LDL-C > 2.8 mmol/L and SGA, with pre-pregnancy BMI, used as an adjustment variable, did not show a linear relationship with log odds, leading to its inclusion as a spline in the model.

Results from linear regression are reported as β-coefficient with 95%CI, and it has been ensured that the data met the assumptions for linear regression. For the linear regression analyses, HbA1c was rescaled to reflect a 5 mmol/mol increase to improve the clarity of the effect estimates.

Post hoc logistic regression between triglyceride and preterm birth, along with triglyceride and NICU > 24 h, was conducted stratified on GDM treatment (insulin or diet-only). Obstetric and perinatal outcomes were compared based on GDM treatment, with BW z-score expressed with β-coefficient [95%CI] and categorical variables as risk ratio (RR) [95%CI]. Characteristic variables from GW 34–38 visit were stratified on treatment and analysed using a standard t-test for continuous variables with normal distribution and equal variance. Data with unequal variance were analysed using Welch's t-test. For continuous variables with non-parametric distribution, the Mann–Whitney (rank-sum) test was used, and for categorical variables, the χ^2-test was used. All tests were two-sided, and the α-level was set to 0.05. No adjustment for multiple comparisons was applied. Analyses were performed using STATA version 18.0 (StataCorp, Texas, USA).

Table 1 Maternal baseline characteristics of the total study population and stratified by GDM treatment (mother/infant pairs = 722)

	Study population ***N*** **= 722**	**Diet-only** ***N*** **= 592 (82.0%)**	**Insulin** ***N*** **= 130 (18.0%)**	***P***
Maternal characteristics	**Mean (SD)/median (IQR) or n (%)**	**Mean (SD)/median (IQR) or n (%)**	**Mean (SD)/median (IQR) or n (%)**	
Age (years)	34.3 (5.0)	34 (5.0)	34 (5.1)	0.96
Missing	0 (0%)	0 (0%)	0 (0%)	
Nullipara	328 (49.4%)	277 (50.5%)	51 (44.3%)	0.23
Missing	58 (8%)	43 (7.3%)	15 (11.5%)	
Pre-pregnancy BMI group	27.8 (6.4)			< 0.001
BMI < 25 kg/m^2	264 (36.7%)	230 (39.0%)	34 (26.2%)	
BMI ≥ 25- < 30 kg/m^2	236 (32.8%)	200 (34.0%)	36 (27.7%)	
BMI ≥ 30 kg/m^2	219 (30.5%)	159 (27.0%)	60 (46.2%)	
Missing	3 (0.42%)	3 (0.51%)	0 (0%)	
Ethnicity				0.005
Europe	509 (76.9%)	433 (79.2%)	76 (66.1%)	
Asia	51 (7.7%)	42 (7.7%)	9 (7.8%)	
Middle east	78 (11.8%)	56 (10.2%)	22 (19.1%)	
Other	24 (3.6%)	16 (2.9%)	8 (7.0%)	
Missing	60 (8.3%)	45 (7.6%)	15 (11.5%)	
Education				0.001
Elementary school	32 (4.9%)	22 (4.0%)	10 (8.8%)	
Short-term education	186 (28.2%)	145 (26.5%)	41 (36.3%)	
Intermediate higher education	232 (35.2%)	196 (35.8%)	36 (31.9%)	
Long higher education	178 (27.0%)	161 (29.4%)	17 (15.0%)	
Other (including foreign education)	32 (4.9%)	23 (4.2%)	9 (8.0%)	
Missing	62 (8.6%)	45 (7.6%)	17 (13.1%)	
Gestational hypertension or PE	66 (9.1%)	44 (7.4%)	22 (16.9%)	0.001
Missing	0 (0%)	0 (0%)	0 (0%)	
GWG in relation to IOM recommendation				< 0.001
Recommended GWG	221 (33.5%)	190 (35.2%)	31 (26.1%)	
High GWG	217 (32.9%)	162 (30.0%)	55 (46.2%)	
Low GWG	221 (33.5%)	188 (34.8%)	33 (27.7%)	
Missing	63 (8.7%)	52 (8.8%)	11 (8.5%)	
Variables related to glycaemia				
GA at GDM diagnose (weeks + days)	28 + 4 (27 + 1–30 + 2)	29 (28–31)	28 (21–29)	< 0.001
Missing	21 (2.9%)	13 (2.2%)	8 (6.2%)	
GDM-diagnostic OGTT 2-h value (mmol/L)	9.8 (9.3–10.7)	9.7 (9.3–10)	11 (9.9–13)	< 0.001
Missing	18 (2.5%)	12 (2%)	6 (4.6)	
Insulin treatment of GDM	130 (18.0%)			
Missing	0 (0%)			
GA at insulin start (weeks + days)	30 + 1 (5 + 3)			
Missing	0 (0%)			
Insulin max dose (IU/kg/day)	0.3 (0.2–0.6)			
Missing	11 (8.5%)			
Biochemical variables measured in GW 34–38				
GA at pregnancy visit with blood samples of biochemical markers (weeks + days)	36 + 6 (1 + 4)	37 (1 + 3)	37 (1 + 4)	0.16
Missing	76 (10.5%)			
Total cholesterol (mmol/L)	6.1 (1.2)	6.1 (1.2)	5.9 (1.1)	0.06
Missing	71 (9.8%)	55 (9.3%)	16 (12.3%)	
Triglyceride (mmol/L)	3.4 (1.2)	3.4 (1.2)	3.5 (1.3)	0.23
Missing	127 (17.6%)	104 (17.6%)	23 (17.7%)	
LDL-C (mmol/L)	2.8 (2.2–3.5)	2.8 (2.3–3.5)	2.6 (2.2–3.3)	0.26
Missing	223 (30.9%)	177 (29.9%)	46 (35.4%)	
LDL-C group				0.29

Table 1 (continued)

	Study population *N* = 722	Diet-only *N* = 592 (82.0%)	Insulin *N* = 130 (18.0%)	*P*
Maternal characteristics	**Mean (SD)/median (IQR) or n (%)**	**Mean (SD)/median (IQR) or n (%)**	**Mean (SD)/median (IQR) or n (%)**	
LDL-C ≤ 2.8 mmol/L	259 (51.9%)	211 (50.8%)	48 (57.1%)	
LDL-C > 2.8 mmol/L	240 (48.1%)	204 (49.2%)	36 (42.9%)	
HbA1c (mmol/mol)	36.2 (4.7)	35 (4.1)	40 (5.1)	< 0.001
Missing	82 (11.4%)	71 (12%)	11 (8.5%)	

Maternal baseline characteristics of the total study population and stratified by GDM treatment (insulin or diet-only) among women with GDM participating in the follow-up program in the Central Denmark Region, April 2019 to December 2022

Continuous variables are presented as mean (SD) if normally distributed and as median (IQR) if non-normally distributed. Comparisons between diet-treated and insulin-treated groups were performed using the standard t-test or Welch's t-test for normally distributed variables, and the Mann–Whitney rank-sum test for non-normally distributed variables

Categorical variables are presented as n (%), and comparisons were performed using the Chi^2 test

Abbreviations: *BMI* body mass index, *PE* preeclampsia, *GWG* gestational weight gain, *IOM* The Institute of Medicine, *GA* gestational age, *GDM* gestational diabetes mellitus, *OGTT* oral glucose tolerance test, *LDL-C* low-density lipoprotein cholesterol, *HbA1c* haemoglobin A1c, *SD* standard deviation, *IQR* interquartile range (25th – 75th percentile)

Results

Characterisation of non-participants

In total, 1270 women participated in the GDM follow-up program. After exclusion of 21 twin pregnancies and 527 cases in which differences in clinical registration practices across hospitals prevented linkage between maternal and infant records, 722 mother–infant pairs were included in the analysis.

The non-eligible individuals ($n = 527$) had a higher pre-gestational BMI, were more often of European descent, fewer had a long higher education, gained less weight in pregnancy, had higher 3rd trimester LDL-C, while having lower HbA1c.

Characteristics and obstetric/perinatal outcomes

The study population's ($n = 722$) characteristics and outcomes are shown in Tables 1 and 2. The diagnostic median OGTT 2-h value was 9.8 (9.3–10.7) mmol/L. Median gestational weight gain was 10.5 (6.7–14.3) kg. According to IOM recommendations, 33.5% had a normal weight gain, 32.9% had a excessive gain, and 33.5% had an insufficient weight gain in pregnancy. The standardised BW z-score was 0.3 (± 1.2), with 20.0% LGA infants and 8.3% SGA infants. Furthermore, 9.7% of infants developed neonatal hypoglycemia within 2 h after birth, and 13.3% of the infants were admitted to the NICU > 24 h (Table 2). Outcomes stratified by GDM treatment (insulin or diet-only) are shown in Fig. 1, and maternal characteristics stratified by GDM treatment are shown in Table 1.

Associations between maternal parameters and obstetric/perinatal outcomes

Crude and adjusted multiple logistic regression analysis for adverse obstetric and perinatal outcomes are shown in Table 3, while linear regression analysis for BW z-score is shown in Table 4. Only associations reaching statistical significance are described here, whereas full model results are presented in Tables 3 and 4. A positive relationship between LGA and almost all exposures investigated was found. A maternal pre-pregnancy BMI of ≥ 30 kg/m2 was positively associated with BW z-score and LGA (adjusted odds ratio (aOR) 2.26 [95%CI 1.37;3.73]) and negatively associated with SGA (aOR 0.37 [95%CI 0.16;0.84]). The diagnostic OGTT 2-h value showed a positive association with BW z-score, LGA (aOR 1.20 [95%CI 1.05;1.38]), and neonatal hypoglycemia. Moreover, 3rd trimester total cholesterol was positively associated with LGA (aOR 1.25 [95%CI 1.05;1.49]), and triglyceride was positively associated with BW z-score and LGA (aOR 1.28 [95%CI 1.07;1.53]). However, triglycerides were negatively associated with preterm birth and admission to NICU > 24 h. LDL-C was associated with a higher BW z-score. HbA1c in the 3rd trimester was positively associated with BW z-score and LGA (aOR 1.13 [95%CI 1.07;1.18]). Lastly, a high GWG was positively associated with BW z-score and LGA (aOR 1.76 [95%CI 1.07;2.87]), while a low GWG was negatively associated with BW z-score and LGA (aOR 0.51 [95%CI 0.29;0.89]) (Tables 3 and 4). Correlation analyses revealed limited associations between glycemic and lipid parameters. A weak positive correlation was observed between HbA1c and triglycerides ($r = 0.16$, $p < 0.001$), while diagnostic OGTT 2-h values showed a weak negative correlation with total cholesterol ($\rho = -0.14$, $p < 0.001$); no other significant correlations were found. Sensitivity analyses adjusting HbA1c models for triglycerides, triglyceride models for HbA1c, OGTT models for total cholesterol, and total cholesterol models for OGTT resulted in only minor changes in aORs.

GDM treatment (insulin or diet-only)

Eighteen percent of the participants were treated with insulin with a maximum daily dose of 30 (16–52) IE,

Table 2 Obstetric and perinatal variables (mother/infant pairs = 722)

Neonatal characteristics	Mean (SD)/Median (IQR) or n (%)
GA at birth (weeks + days)	39 + 3 (38 + 3—40 + 3)
Missing	0 (0%)
Birthweight (g)	3539 (516.5)
Missing	0 (0%)
Female sex	343 (47.6%)
Missing	2 (0.3%)
Method of delivery	
Vaginal, spontaneous	494 (68.7%)
Vaginal, assisted	40 (5.6%)
Emergency C-section	102 (14.2%)
Elective C-section	83 (11.5%)
Missing	3 (0.4%)
Outcomes	
Preterm birth	43 (6.0%)
Missing	0 (0%)
Shoulder dystocia	16 (2.2%)
Missing	0 (0%)
Birthweight z-score	0.3 (1.2)
Missing	3 (0.4%)
LGA	144 (20.0%)
Missing	3 (0.4%)
SGA	60 (8.3%)
Missing	3 (0.4%)
Apgar score 5 min≤7	15 (2.1%)
Missing	1 (0.1%)
Neonatal hypoglycemia	67 (9.7%)
Missing	29 (4%)
Jaundice requiring phototherapy	52 (7.2%)
Missing	1 (0.1%)
RDS	18 (2.5%)
Missing	1 (0.1%)
NICU > 24 h	96 (13.3%)
Missing	1 (0.1%)

Neonatal characteristics and obstetric outcomes of infants born to women with GDM participating in the follow-up program in the Central Denmark Region, April 2019 to December 2022

Continuous variables normally distributed are summarised with mean (SD) and variables not normally distributed are summarised with median (IQR)

Categorical variables are summarised with n (%)

Abbreviations: *GA* gestational age, *C-section* caesarean section, *LGA* large-for-gestational-age, *SGA* small-for-gestational-age, *RDS* respiratory distress syndrome, *NICU > 24 h* admission to the neonatal intensive care unit more than 24 h, *SD* standard deviation, *IQR* interquartile range (25th – 75th percentile)

corresponding to 0.3 (0.2–0.6) IE/kg/day, based on the weight from the GW 34–38 visit. Fifteen percent of Europeans, 18% of Asians, and 28% of Middle Easterners were insulin-treated. The insulin-treated group differed from the diet-only treated group in that fewer participants identified as of European ethnicity, fewer had intermediate or long-term education, and more participants were in the pre-pregnancy BMI ≥ 30 kg/m^2 category. Additionally, there were more cases of high GWG, earlier GDM diagnosis with a higher diagnostic OGTT 2-h value, and increased HbA1c levels in the third trimester (Table 1). The insulin-treated group had increased RR for LGA (RR 2.51 [95%CI 1.89;3.33]) and gestational hypertension/PE (RR 2.28 [95%CI 1.41;3.66]) (Fig. 1). Furthermore, the mean BW z-score was significantly different between the two groups ($p < 0.001$ with a standard t-test). The post hoc logistic regression between triglyceride and preterm birth, and between triglyceride and admission to NICU > 24 h stratified by GDM treatment (insulin or diet-only), could only obtain crude estimates, due to a few events in the insulin-treated group. Here, triglyceride remained negatively associated with preterm birth (crude odds ratio (cOR) 0.56 [95%CI 0.34;0.91]) and admission to NICU > 24 h (cOR 0.67 [95%CI 0.52;0.91]) in the diet-only treated group.

Discussion

This study aimed to broaden the understanding of GDM beyond its glycemic focus. Higher 3rd trimester total cholesterol and triglyceride levels were found to be positively associated with LGA infants, and triglyceride was also associated with higher BW z-score. Furthermore, higher pre-pregnancy BMI and high GWG raised the odds of high BW z-score and LGA, while low GWG reduced this risk. Moreover, insulin-treated women faced higher risks of LGA and gestational hypertension/PE, than those managed by diet-only.

Lipids associated with birth weight z-score and LGA

Third-trimester total cholesterol and triglycerides were positively associated with LGA infants in pregnancies complicated by GDM. This highlights maternal lipid metabolism as a potential contributor to adverse neonatal outcomes, strengthening the idea of GDM-related risks extending beyond hyperglycemia [8, 24–27]. Our results support previous studies linking dysregulated maternal lipid metabolism to accelerated fetal growth, especially in GDM pregnancies, which are already under increased metabolic stress [8, 25]. Findings are further in alignment with Barbour et al., who found that postprandial triglyceride levels predict newborn adiposity more strongly than glucose levels in women with obesity but not necessarily GDM [14]. Several mechanisms could explain the association. Increased maternal triglyceride levels are thought to enhance the transfer of nutrients across the placenta, particularly free fatty acids, promoting fetal adiposity and growth, though the exact mechanism is not fully understood yet [24, 27, 28].

The current findings could have important clinical implications. While glycemic control is fundamental in managing GDM, our results suggest that maternal lipid profiles could be considered in the risk assessment of adverse outcomes, such as LGA, which is in line with other studies [15]. The correlation analyses revealed

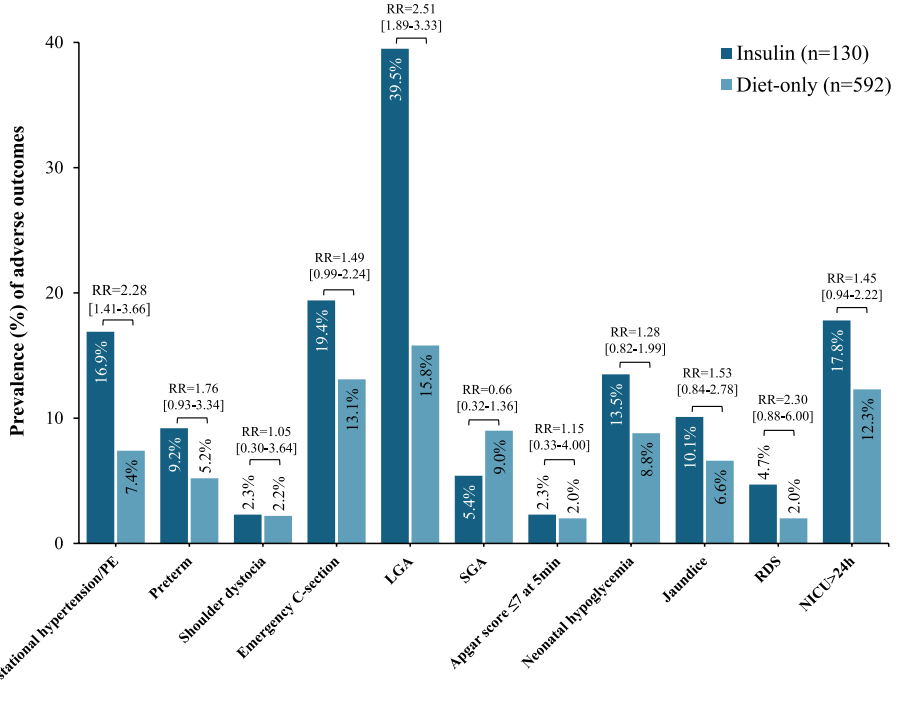


Fig. 1 Adverse outcomes by treatment group—prevalence (%) and risk ratio (RR) with 95% confidence interval [95%CI] of adverse outcomes between treatment groups (insulin vs. diet-only). *Abbreviations:* PE = preeclampsia. C-section = cesarean section. LGA = large-for-gestational-age. SGA = small-for-gestational-age. RDS = respiratory distress syndrome. NICU > 24 h = admission to the neonatal intensive care unit longer than 24 h

limited associations between glycemic and lipid parameters, suggesting that these markers reflect partially distinct metabolic features in GDM pregnancies, with only small mutual confounding. Current clinical practices often overlook lipid screening in GDM, potentially missing the chance to identify mother-infant pairs at a higher risk for complications. On the other hand, there are only limited literature reports on the effects of addressing lipid metabolism in relation to fetal growth and neonatal complications. In this respect, the dietary guidance that is the mainstay in treating GDM is usually centred around the intake of carbohydrates and overall energy intake. Less effort is put into the guidance to modify maternal lipid levels.

Lipids and other adverse outcomes

Surprisingly, our results showed that higher 3rd trimester triglyceride levels were negatively associated with both preterm birth and admission to NICU > 24 h, the latter probably as a result of the association between preterm birth and admission to NICU > 24 h. Recent research on lipids and GDM complications has mainly focused on LGA and macrosomia, so to our knowledge, not many studies have explored the relationship between 3rd trimester total cholesterol, triglyceride, LDL-C, and the outcomes of preterm birth (and NICU admissions) in a GDM cohort. However, a study by Shi et al. conducted between 2011–2021 investigated maternal lipid profiles in the 2nd and 3rd trimesters and the relationship to perinatal outcomes in a Chinese cohort of 1632 mother-infant pairs with GDM and 9067 non-GDM. In contrast to our study, Shi et al. found that for the 3rd trimester lipid profile of women with GDM, higher triglyceride levels were positively associated with NICU admission and C-section but found no association between triglyceride and preterm birth [29]. However, a direct comparison is challenging because our study population may represent a more severely affected GDM group based on differing diagnostic and exclusion criteria. A possible explanation for the negative association between 3rd trimester triglyceride levels with preterm birth and admission to NICU > 24 h could be the insulin therapy among women with the most severe GDM, as low triglyceride levels could be a result of insulin therapy. This explanation seems plausible considering the increased complication rate among insulin-treated patients (Fig. 1). To accommodate this consideration, stratification based on GDM treatment (insulin or diet-only) was performed, but due to few events in the insulin-treated group, only crude estimates could be made. Both groups showed the same negative trend, and the absence of statistical significance in the insulin-treated group likely reflects the limited statistical power due to the smaller sample size in this group ($n = 130$; 12 preterm) versus the diet-only group ($n = 592$; 31 preterm). Larger studies with a bigger sample

Table 3 Association between maternal factors and adverse outcomes

	Preterm birth	LGA	SGA	Neonatal hypoglycaemia	NICU > 24 h
Pre-pregnancy BMI group					
BMI < 25 kg/m^2	Ref	Ref	Ref	Ref	Ref
BMI ≥ 25 to < 30 kg/m^2					
cOR [95%CI]	1.30 [0.62;2.72]	1.65 [1.03;2.64]	0.79 [0.43;1.42]	0.96 [0.53;1.76]	1.14 [0.66;1.94]
aOR[1] [95%CI]	1.14 [0.53;2.47]	1.63 [1.00;2.72]	0.86 [0.46;1.60]	1.01 [0.54;1.91]	1.12 [0.63;1.99]
BMI ≥ 30 kg/m^2					
cOR [95%CI]	1.13 [0.52;2.51]	2.33 [1.47;3.70]	0.35 [0.16;0.75]	0.95 [0.51;1.77]	1.54 [0.91;2.60]
aOR[1] [95%CI]	0.99 [0.44;2.26]	2.26 [1.37;3.73]	0.37 [0.16;0.84]	0.98 [0.51;1.90]	1.61 [0.93;2.79]
OGTT 2-h value (mmol/L)					
cOR [95%CI]	1.20 [1.01;1.43]	1.19 [1.05;1.34]	0.80 [0.62;1.03]	1.25 [1.08;1.44]	1.16 [1.01;1.32]
aOR[2] [95%CI]	1.23 [1.01;1.47]	1.20 [1.05;1.38]	0.80 [0.62;1.04]	1.30 [1.10;1.52]	1.15 [1.00;1.34]
Total cholesterol (mmol/L)					
cOR [95%CI]	0.72 [0.50;1.03]	1.19 [1.01;1.40]	0.90 [0.71;1.15]	0.94 [0.74;1.19]	0.80 [0.65;1.00]
aOR[2] [95%CI]	0.66 [0.45;0.97]	1.25 [1.05;1.49]	0.81 [0.63;1.06]	0.97 [0.76;1.24]	0.78 [0.63;0.98]
Triglyceride (mmol/L)					
cOR [95%CI]	0.60 [0.40;0.91]	1.27 [1.08;1.49]	1.00 [0.80;1.26]	0.80 [0.62;1.05]	0.69 [0.55;0.88]
aOR[2] [95%CI]	0.59 [0.38;0.90]	1.28 [1.07;1.53]	0.93 [0.72;1.20]	0.84 [0.64;1.11]	0.64 [0.50;0.83]
LDL-cholesterol (mmol/L)					
≤ 2.8 mmol/L	Ref	Ref	Ref	Ref	Ref
> 2.8 mmol/L					
cOR [95%CI]	1.30 [0.56;3.09]	1.36 [0.86;2.16]	0.84 [0.45;1.58]	0.84 [0.46;1.54]	0.80 [0.48;1.34]
aOR[2] [95%CI]	1.25 [0.52;3.03]	1.47 [0.90;2.40]	0.75 [0.38;1.49]*	0.95 [0.51;1.77]	0.82 [0.48;1.40]
HbA1c (mmol/mol)					
cOR [95%CI]	1.01 [0.94;1.10]	1.14 [1.09;1.19]	0.90 [0.85;0.96]	1.02 [0.97;1.08]	1.03 [0.98;1.08]
aOR[2] [95%CI]	1.03 [0.94;1.12]	1.13 [1.07;1.18]	0.88 [0.82;0.95]	1.01 [0.95;1.08]	1.04 [0.98;1.01]
GWG (IOM guideline)					
Recommended GWG	Ref	Ref	Ref	Ref	Ref
High GWG					
cOR [95%CI]	1.14 [0.45;2.86]	1.69 [1.08;2.64]	0.70 [0.32;1.49]	0.60 [0.31;1.17]	1.22 [0.68;2.17]
aOR[1] [95%CI]	1.10 [0.43;2.79]	1.76 [1.07;2.87]	0.61 [0.27;1.42]	0.63 [0.32;1.25]	1.22 [0.66;2.24]
Low GWG					
cOR [95%CI]	1.84 [0.80;4.25]	0.54 [0.32;0.92]	1.59 [0.84–3.03]	0.72 [0.38;1.37]	1.50 [0.86;2.63]
aOR[1] [95%CI]	1.48 [0.62;3.54]	0.51 [0.29;0.89]	1.86 [0.94–3.69]	0.66 [0.34;1.28]	1.40 [0.77;2.54]

Crude and adjusted logistic regression analyses of maternal characteristics and adverse obstetric/perinatal outcomes in women with GDM participating in the follow-up program in the Central Denmark Region, April 2019 to December 2022

Abbreviations: *BMI* Body Mass Index, *OGTT* oral glucose tolerance test, *LDL-C* low-density lipoprotein cholesterol, *HbA1c* haemoglobin A1c, *GWG* gestational weight gain, *IOM* The Institute of Medicine, *cOR* crude odds ratio, *aOR[1]* odds ratio adjusted for maternal age (years), ethnicity (group), and parity (group), *aOR[2]* odds ratio adjusted as aOR[1] + pregestational BMI (kg/m2), *[95%CI]* 95% confidence interval

*The assumption of a linear relationship between the log-odds and the variable 'pregestational BMI' (which the model adjusts for) is not met. Therefore, 'pregestational BMI' is included in this model as a spline to better capture its relationship with the outcome

size of insulin-treated patients are needed to confirm the observed association.

Pre-pregnancy BMI and gestational weight gain

BMI ≥ 30 kg/m^2 was positively associated with BW z-score and LGA, while negatively associated with SGA. GWG significantly influenced the BW z-score and LGA risk; high GWG was positively associated with BW z-score and LGA, whereas low GWG was negatively associated with both. These findings are in line with other studies [12, 30]. Differentiating the effects of GDM and obesity is challenging, with conflicting findings in the literature [31, 32]. However, evidence, including the HAPO study sub-analyses, shows predominantly that both GDM and obesity are independent risk factors for a range of adverse pregnancy outcomes, and the combined effect exceeds each condition's impact separately [13].

Maternal glycemic status; 3rd trimester HbA1c and diagnostic OGTT 2-h value

Our study confirms that maternal glycemic status, assessed through diagnostic OGTT 2-h value and 3rd trimester HbA1c, affects the risk of adverse outcomes, especially LGA. High diagnostic OGTT values are well known to be associated with adverse outcomes [5]. Studies investigating 3rd trimester HbA1c association with adverse outcomes

Table 4 Association between maternal factors and birth weight z-score

	Birth weight z-score
Pre-pregnancy BMI group	
BMI < 25 kg/m^2	Ref
BMI ≥ 25—< 30 kg/m^2	
cβ-coef. [95%CI]	0.27 [0.06;0.47]
aβ-coef.^{1} [95%CI]	0.25 [0.04;0.45]
BMI ≥ 30 kg/m^2	
cβ-coef. [95%CI]	0.54 [0.33;0.75]
aβ-coef.^{1} [95%CI]	0.50 [0.29;0.71]
OGTT-value (mmol/L)	
cβ-coef. [95%CI]	0.12 [0.06;0.18]
aβ-coef.^{2} [95%CI]	0.12 [0.06;0.19]
Total cholesterol (mmol/L)	
cβ-coef. [95%CI]	0.08 [0.00;0.16]
aβ-coef.^{2} [95%CI]	0.11 [0.03;0.18]
Triglyceride (mmol/L)	
cβ-coef. [95%CI]	0.11 [0.03;0.18]
aβ-coef.^{2} [95%CI]	0.13 [0.05;0.21]
LDL-C (mmol/L)	
≤ 2.8 mmol/L	Ref
> 2.8 mmol/L	
cβ-coef. [95%CI]	0.21 [0.01;0.42]
aβ-coef.^{2} [95%CI]	0.25 [0.05;0.46]
HbA1c (5 mmol/mol)	
cβ-coef. [95%CI]	0.37 [0.28;0.46]
aβ-coef.^{2} [95%CI]	0.37 [0.26;0.47]
GWG (IOM recommendation)	
Recommended GWG	Ref
High GWG	
cβ-coef. [95%CI]	0.38 [0.16;0.59]
aβ-coef.^{1} [95%CI]	0.42 [0.21;0.63]
Low GWG	
cβ-coef. [95%CI]	−0.24 [−0.46;−0.03]
aβ-coef.^{1} [95%CI]	−0.27 [−0.48;−0.06]

Linear regression analyses of maternal characteristics and birth weight z-score in women with GDM participating in the follow-up program in the Central Denmark Region, April 2019 to December 2022

Abbreviations: *BMI* Body Mass Index, *OGTT* oral glucose tolerance test, *LDL-C* low-density lipoprotein cholesterol, *HbA1c* haemoglobin A1c, *GWG* gestational weight gain, *IOM* The Institute of Medicine, *cβ-coef.* crude β-coefficient, *[95%CI]* 95% confidence interval, *aβ-coef.*^{1}β-coefficient adjusted for maternal age (years), ethnicity (group), and parity (group), *aβ-coef.*^{2}β-coefficient adjusted as aβ-coef.^{1} + pregestational BMI (kg/m2)

in GDM pregnancies are limited. However, our findings align with others and add to the evidence of an association between 3rd trimester HbA1c levels and LGA [6, 7]. A higher OR for the diagnostic OGTT 2-h value was observed for BW z-score and LGA, compared to HbA1c. This suggests the diagnostic OGTT 2-h value at a median GA 28 + 4 weeks is more strongly associated with adverse outcomes than HbA1c levels at a mean GA 36 + 6 weeks. Interestingly, studies focusing on mid-term HbA1c provide more consistent evidence of associations with adverse outcomes. For instance, Lowe et al. found that HbA1c from GW 24–32 in uncomplicated pregnancies was associated with outcomes like preterm delivery, primary C-section, BW > 90th percentile, and neonatal hypoglycemia [5]. Lowe et al. also noted that diagnostic OGTT values (fasting, 1-h, and 2-h) showed stronger associations with adverse outcomes, particularly neonatal anthropometry such as BW > 90th percentile [5]. A possible explanation is that HbA1c reflects average blood glucose over weeks, smoothing out short but clinically relevant blood glucose spikes [33]. Meanwhile, the OGTT value is the response to a glucose load, reflecting a postprandial blood glucose response. A high diagnostic OGTT value likely mirrors higher postprandial glucose levels during pregnancy. These glucose spikes may cause fetal hyperinsulinemia and potentially drive macrosomia in these infants [33]. However, a 3rd trimester HbA1c could remain helpful in aiding clinical decisions such as commencing and dosing insulin therapy and planning delivery.

Insulin treatment as a marker for increased pregnancy risks

Insulin-treated women had around a two-and-a-half-fold increased risk of having an LGA infant and a greater risk of gestational hypertension/PE compared to diet-only treated women (Fig. 1).

These findings align with those of other studies, such as Karkia et al.'s recent large cohort study conducted in the UK, which reported similar population characteristics and insulin treatment rates [34]. In general, insulin-treated women have hyperglycemia that lifestyle and diet intervention alone cannot control, leading to prolonged elevated plasma glucose levels. The increased rate of adverse outcomes in this group indicates that optimal treatment has not yet been achieved. Possible reasons for this include delayed insulin initiation, insufficient dose escalation, or specific metabolic characteristics of these mother-infant pairs requiring further intervention.

Strengths and limitations

The strengths of this study include access to detailed information through medical records, minimising missing data on mother-infant pairs, a relatively large population with equal healthcare access, and an observational design that provides a real-world setting, making the results more generalisable to a clinical context.

There are, however, also limitations. Mother-infant pairs not eligible for the study were primarily from regional hospitals, while those included mainly lived in the university hospital's catchment area. Consequently, the study population mostly represents an urban demography with greater ethnic diversity and higher education than ineligible mother-infant pairs. While this increases the risk of selection bias and limits the generalisability of the findings to other populations, adjustments for ethnicity and education level in the analysis help reduce these factors' impact. Additionally, the majority of women with GDM needing insulin treatment were

referred to the university hospital, potentially introducing another selection bias. That said, the proportion receiving insulin in the cohort is comparable to other studies. While insulin therapy is generally considered to impact lipid metabolism, the question of whether it should influence the interpretation of lipid parameters and perinatal outcomes is raised. However, it remains to be explored whether the low doses of insulin administered in this cohort had a clinically relevant effect on the women's lipid profiles.

Non-fasting blood samples in GW 34–38 could impact the comparison with some studies. However, no current consensus exists on whether fasting or postprandial analyte levels are better predictors of fetal overgrowth [14, 35]. Several studies and guidelines support that non-fasting lipid levels can be used as well as fasting lipid levels in a clinical setting for practical and safety reasons, especially for screening [36, 37]. This is supported by the pilot study conducted by Barrett et al., who found no difference in capillary triglyceride concentrations between fasting and 2-h postprandial levels in late pregnancy [35].

Furthermore, the diagnostic criteria for GDM may differ between countries, making generalisations more difficult. Some studies use WHO or IADPSG criteria for OGTT interpretations, while others follow local/national guidelines. This study applied Danish guidelines, including only a OGTT 2-h measure with a 9.0 mmol/L threshold in pregnant women with risk factors in GW 24–28. Compared to e.g. the WHO recommendation of a 2-h value of 8.5 mmol/L defining GDM, our population may therefore display a more severe GDM.

An additional limitation is the different degree of missing data in exposure variables, resulting in varying sample sizes between analyses. Variables with more missing data, such as LDL-C, may have had reduced statistical power, which could have decreased the likelihood of finding statistically significant associations. Accordingly, non-significant findings should be interpreted with caution.

Finally, statistical limitations include that linear regression estimates mean differences in BW z-score, and potential variation in associations across the outcome distribution may not have been captured. Future studies applying quantile regression may provide additional insight into such heterogeneity. Furthermore, it should be noted that these analyses were exploratory, and no adjustment was made for multiple comparisons.

Conclusions and perspectives

Our results highlight the potential of a nuanced risk assessment in GDM pregnancies. The study of 722 mother-infant pairs with maternal GDM found that the lipid profiles, along with glycemic and weight status, were associated with adverse outcomes. Furthermore, insulin-treated women had an increased risk of giving birth to an LGA infant compared to diet-only treated women. These findings advocate for a broader clinical focus beyond glycemic control, potentially including lipid screening, weight, and treatment status in the risk stratification for adverse outcomes.

Abbreviations

BMI	Body mass index
GDM	Gestational diabetes mellitus
OGTT	Oral glucose tolerance test
PE	Preeclampsia
GWG	Gestational weight gain
LDL-C	Low-density lipoprotein cholesterol
HbA1c	Haemoglobin A1c
C-section	Caesarean section
GW	Gestational week
GA	Gestational age
BW	Birth weight
LGA	Large-for-gestational-age
SGA	Small-for-gestational-age
RDS	Respiratory distress syndrome
NICU	Neonatal intensive care unit
NICU > 24 h	Admission to the neonatal intensive care unit lasting longer than 24 h
SD	Standard deviation
IQR	Interquartile range
OR	Odds ratio
aOR	Adjusted odds ratio
cOR	Crude odds ratio

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Authors' contributions

UK, PGO, JF, SK, LRM and HH jointly conceived and designed the study, with SK and MLM providing additional support for statistical analyses. HH collected and curated additional data, performed data cleaning and statistical analyses, and drafted the first version of the manuscript. PRBG and ANS assisted HH in data collection. LLK contributed to data cleaning and data analysis. All authors contributed to the interpretation of results, revised the manuscript, and approved the final version.

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Data availability

The data underlying this study are not publicly available because they contain sensitive participant information and are subject to legal restrictions. The analysis code is publicly available at Zenodo (DOI:10.5281/zenodo.19559696) [38].

Declarations

Ethics approval and consent to participate

Given the universal introduction of the follow-up program, ethical approval was not required according to the Danish data protection legislation. However, the women gave informed consent to participate, and approval for a retrospective audit of the electronic medical records was obtained (Region Midtjylland—VID number 1–45–70–111–23).

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

Author details
[1]Steno Diabetes Center Aarhus, Aarhus University Hospital, Aarhus, Denmark
[2]Department of Clinical Medicine, Aarhus University, Aarhus, Denmark
[3]Danish Diabetes and Endocrine Academy, Odense University Hospital, Odense, Denmark
[4]Department of Internal Medicine, Gødstrup Hospital, Gødstrup, Denmark
[5]Department of Internal Medicine, Horsens Regional Hospital, Horsens, Denmark
[6]Department of Internal Medicine, Randers Regional Hospital, Randers, Denmark
[7]Medical Diagnostic Center, University Clinic for Innovative Patient Pathways, Regional Hospital Central Jutland, Viborg, Denmark
[8]Department of Gynaecology and Obstetrics, Aarhus University Hospital, Aarhus, Denmark

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